

Uber Supply-Demand Gap Analysis Project

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Tools Used: Python, SQL, Excel

Dataset: Uber request data

Project Overview:

This project explores Uber's request data to uncover the reasons behind unfulfilled trip requests. By conducting Exploratory Data Analysis (EDA), writing SQL queries, and building Excel dashboards, we aim to understand when, where, and why Uber experiences cancellations and car unavailability.

Problem Statement:

This Uber frequently encounters service failures caused by either driver cancellations or no available cars, especially during peak hours.

Objective: Identify patterns behind these issues and suggest actionable improvements to enhance trip fulfillment rates, driver allocation, and customer experience.

Business Objective:

The primary objective of this project is to help Uber identify and address the mismatch between ride demand and driver availability. By analyzing historical trip data, we aim to uncover when and where these mismatches occur—specifically during peak demand periods or underserved locations—and understand the reasons behind ride cancellations and unavailability of cars.

This analysis will support Uber in:

- Reducing unfulfilled trip requests caused by either driver cancellations or lack of available vehicles.
- Optimizing driver deployment strategies across different time slots and pickup points.
- Enhancing customer satisfaction by improving ride availability during high-demand periods.
- Supporting data-driven decisions for incentives, operations, and fleet management to maximize trip fulfillment.

Ultimately, the goal is to minimize supply-demand gaps, increase operational efficiency, and drive positive business impact through actionable insights from the data.

Insights from the Analysis

1. Trip Fulfillment Overview

Only 56.5% of all ride requests were successfully fulfilled.

Around 43.5% of requests remained unfulfilled due to either cancellations or no cars being available.

2. Time Slot Analysis

Evening (5 PM–9 PM) had the highest number of ride requests but also a high gap rate.

Late Night (12 AM–5 AM) showed very low ride fulfillment, primarily due to no cars available.

Early Morning (5 AM–8 AM) had the highest number of cancellations, indicating driver-side issues.

3. Pickup Point Behavior

Airport showed a higher percentage of unfulfilled trips compared to the City.

Most unfulfilled Airport trips occurred during Late Night, suggesting low driver availability during those hours.

4. Trip Status Breakdown

Trip Completed: Mostly happens between 7 AM and 10 PM.

Cancelled: Spikes in Early Morning, possibly due to low driver motivation or distance concerns.

No Cars Available: Common during Late Night and Evening, implying a supply shortage.

5. Hourly Request Patterns

Two demand peaks observed:

- Morning Peak: 8–10 AM
- Evening Peak: 5–9 PM

Least requests were seen between 1 AM and 4 AM.

6. Heatmap Findings

The worst service gaps were seen during Late Night at Airport pickups.

The City showed better performance overall, but still suffered during high-demand times.

7. Status by Pickup Point

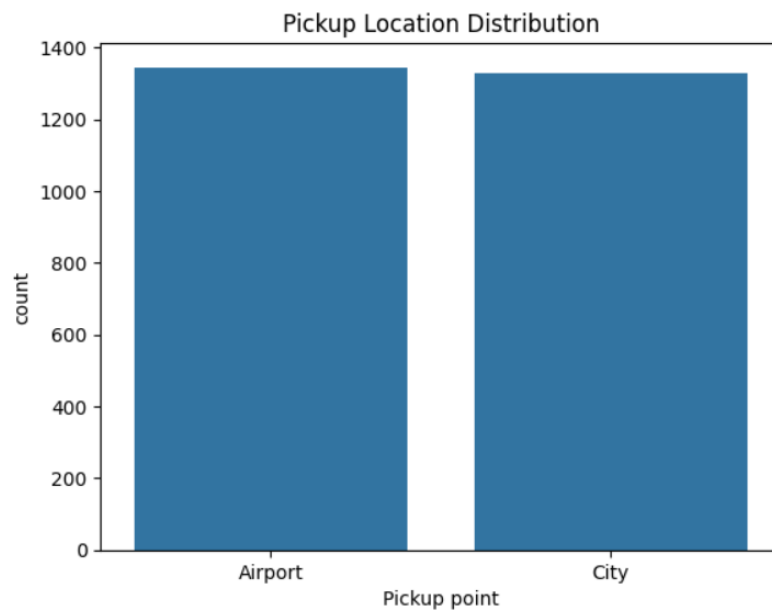
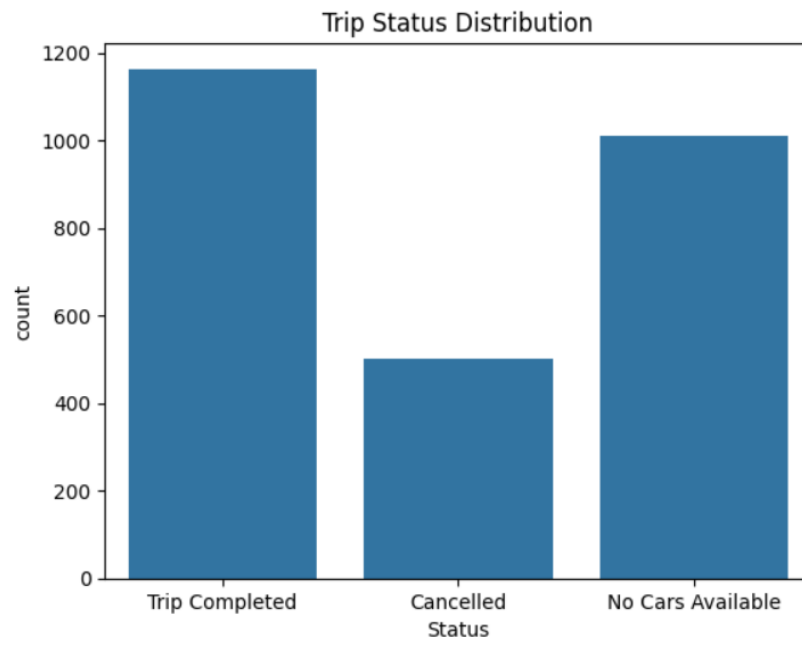
City: More total requests, higher completion rate.

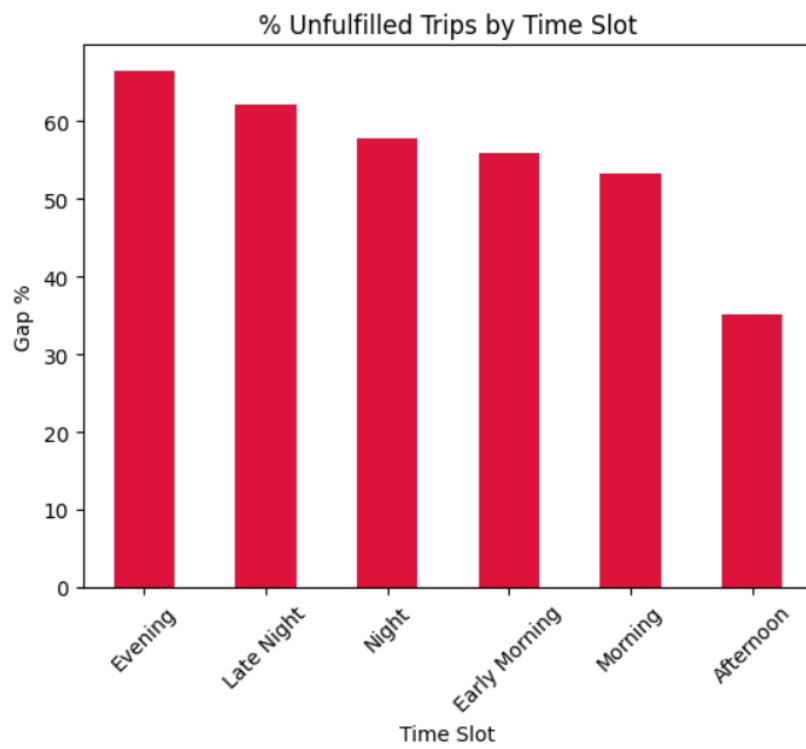
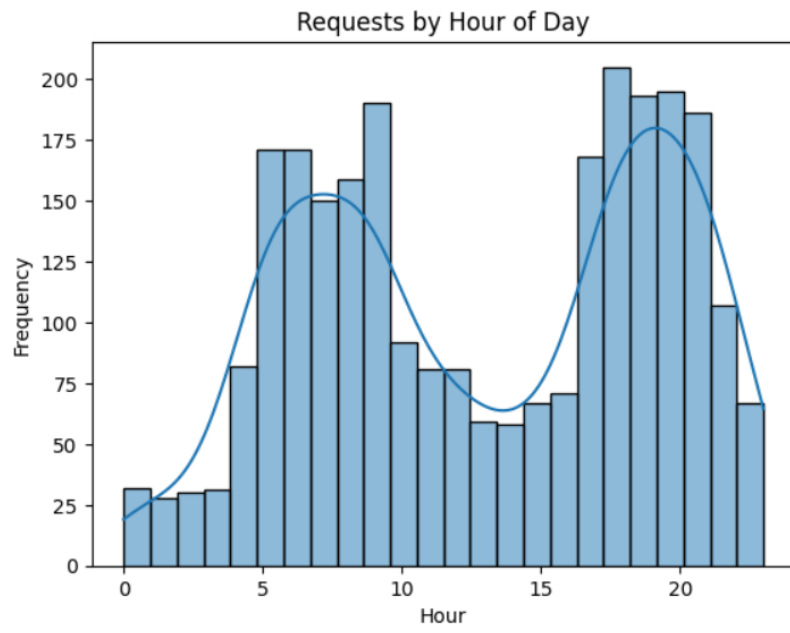
Airport: Fewer total requests but lower completion rate due to time-specific supply issues.

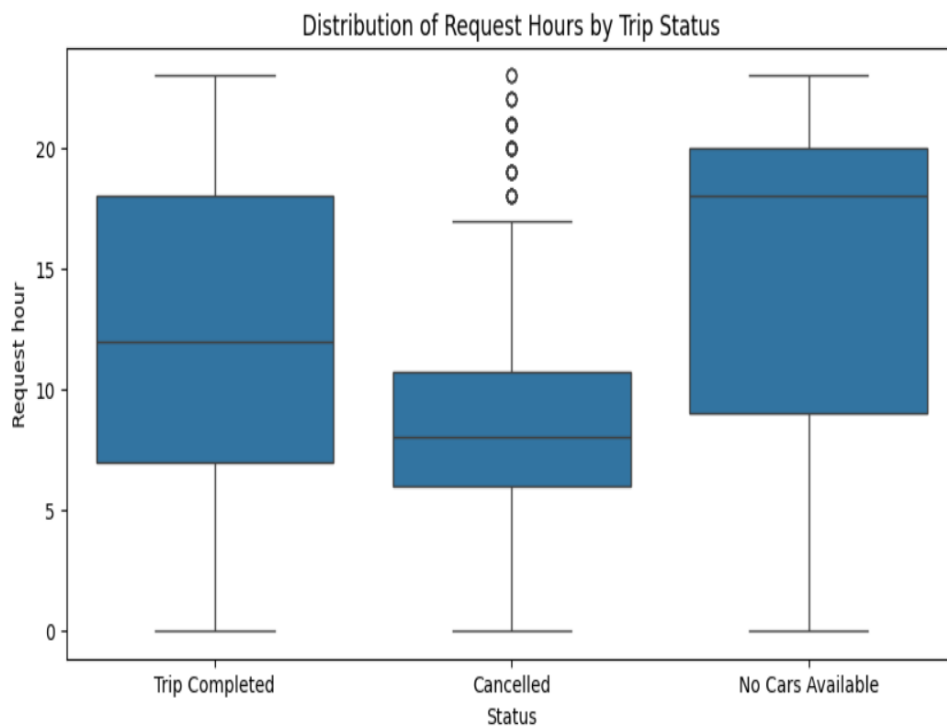
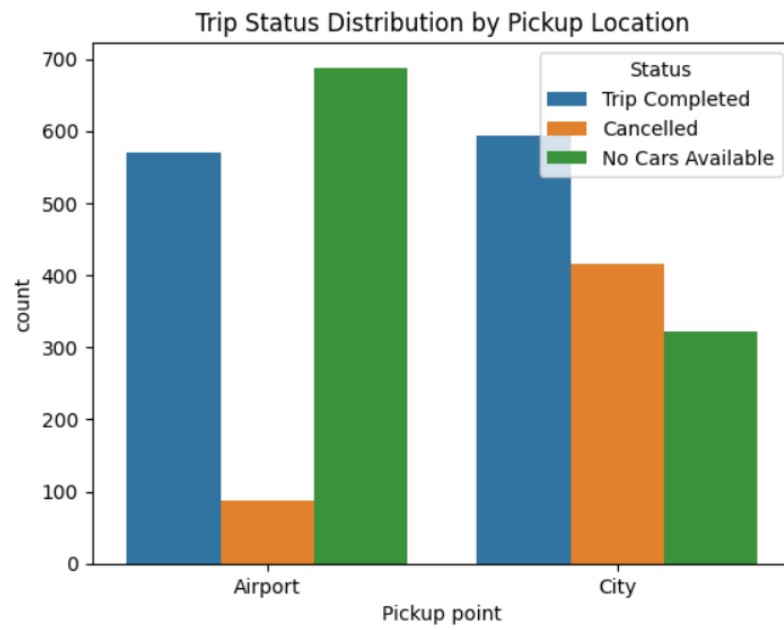
8. SQL & Excel Dashboard Confirmation

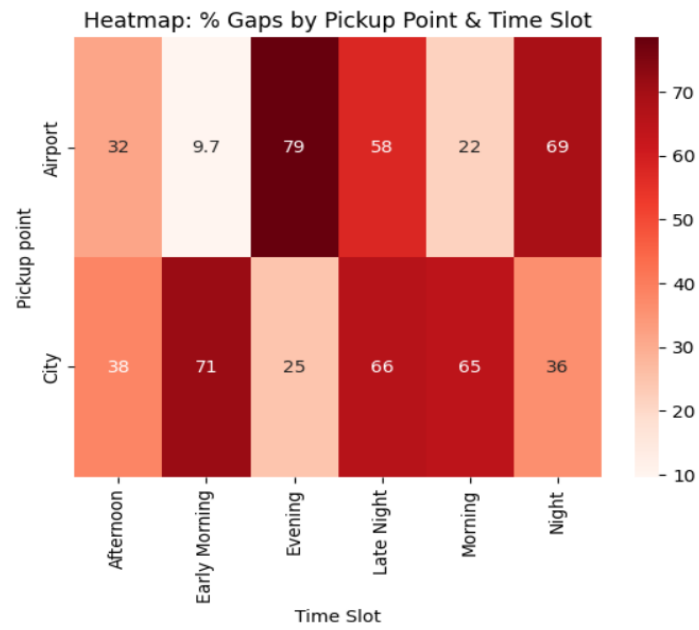
All trends in Python visualizations were validated using SQL queries and Excel dashboards.

Excel dashboard showed visually strong evidence of Evening demand and Early Morning cancellations.

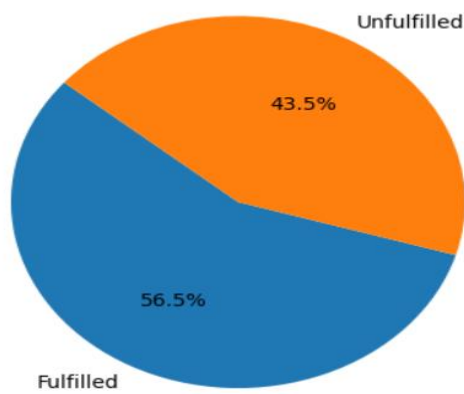




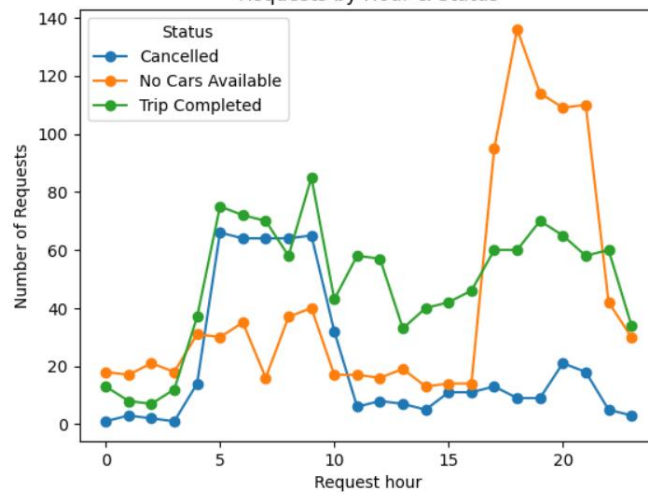


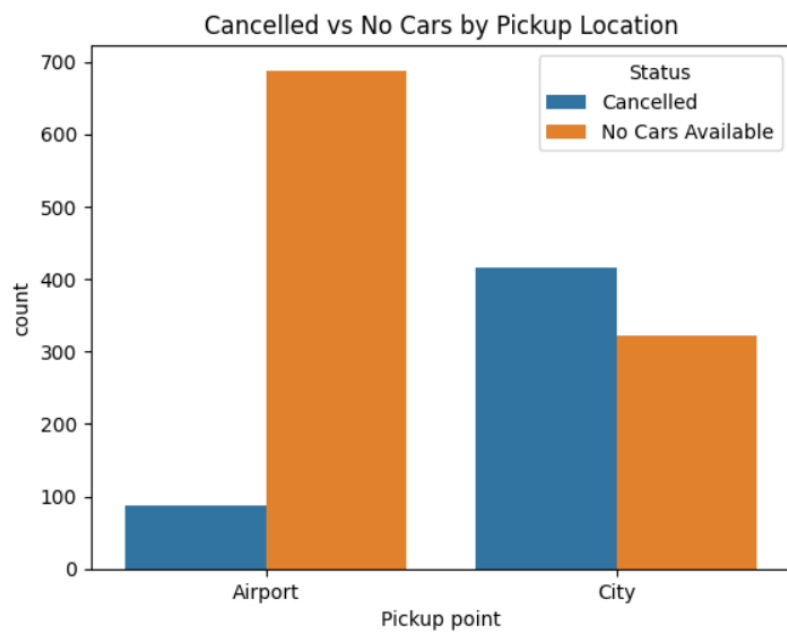
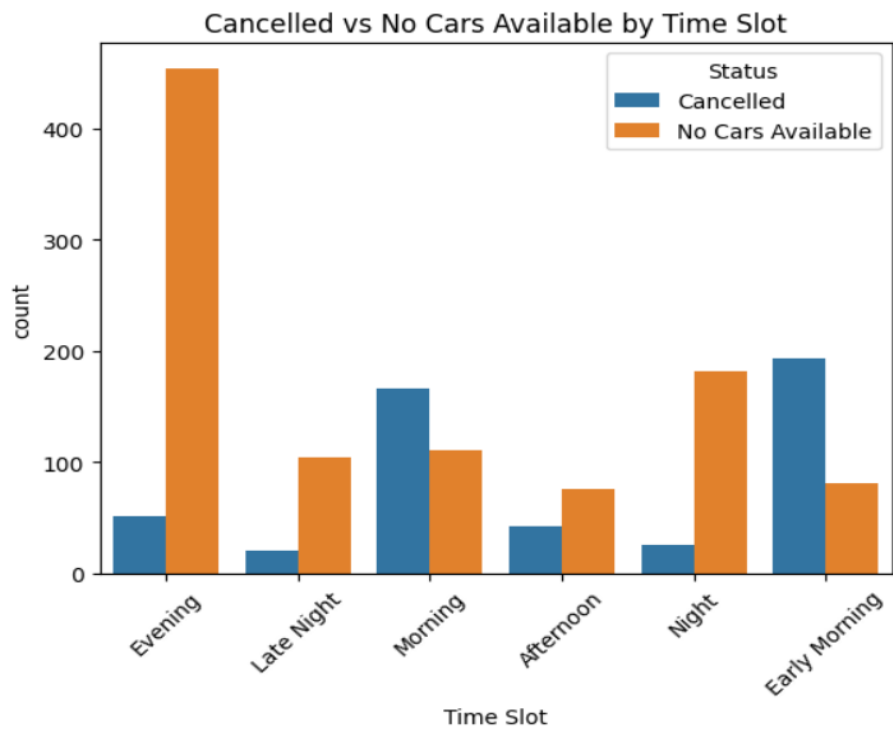


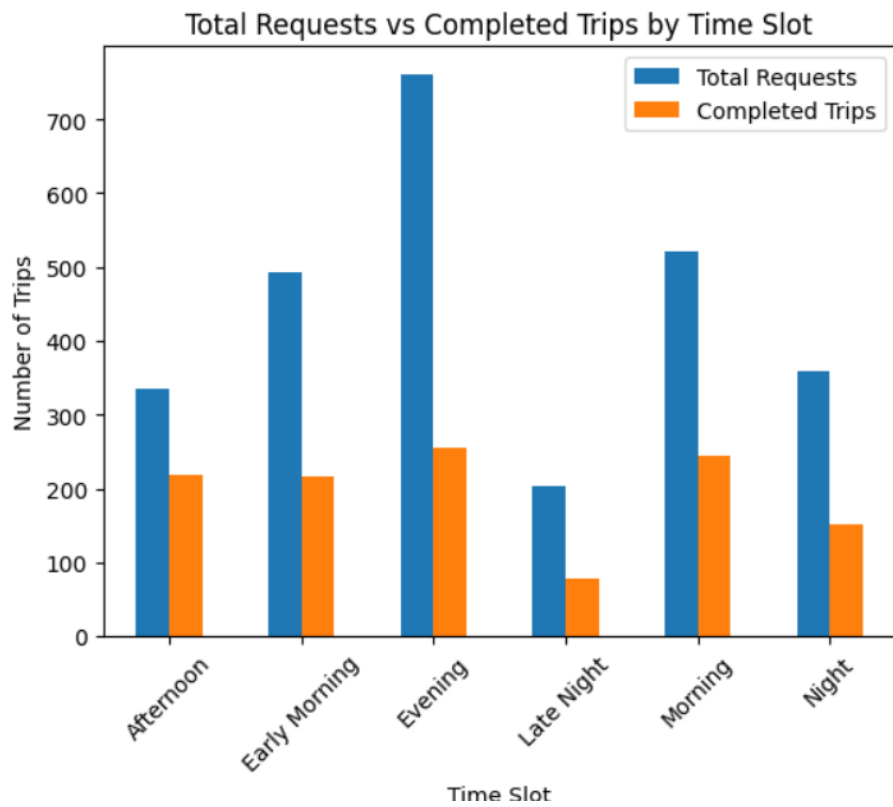
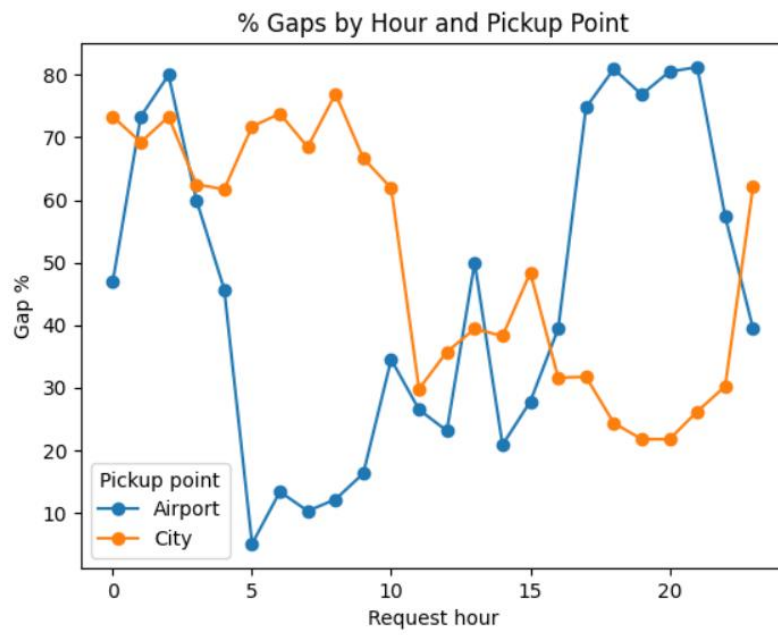
Overall Trip Fulfillment Ratio

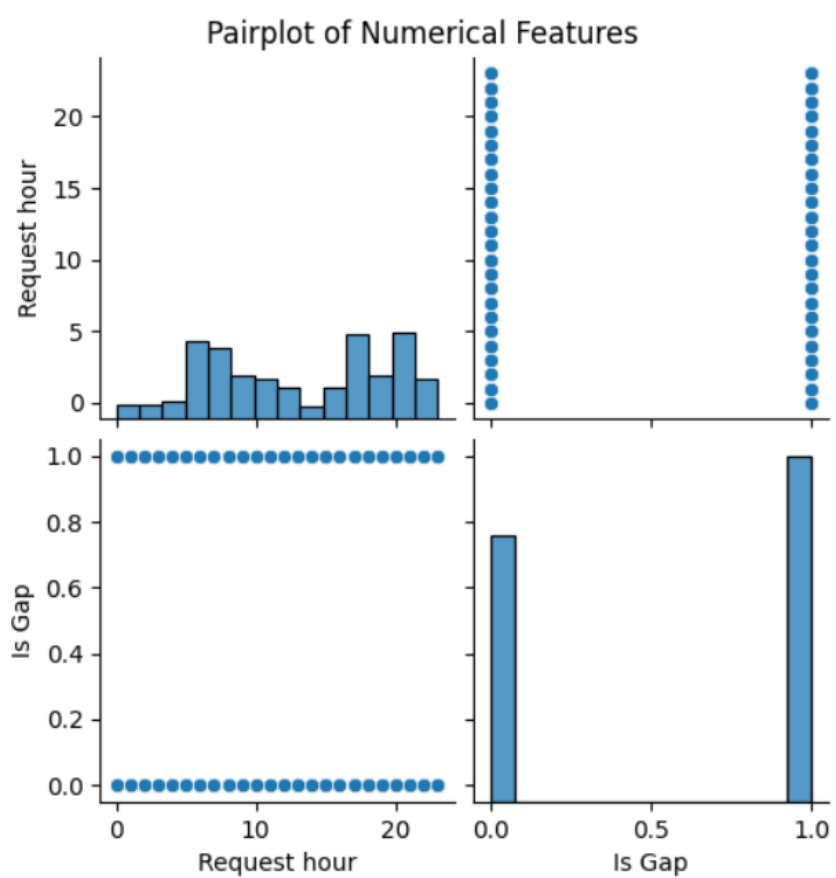
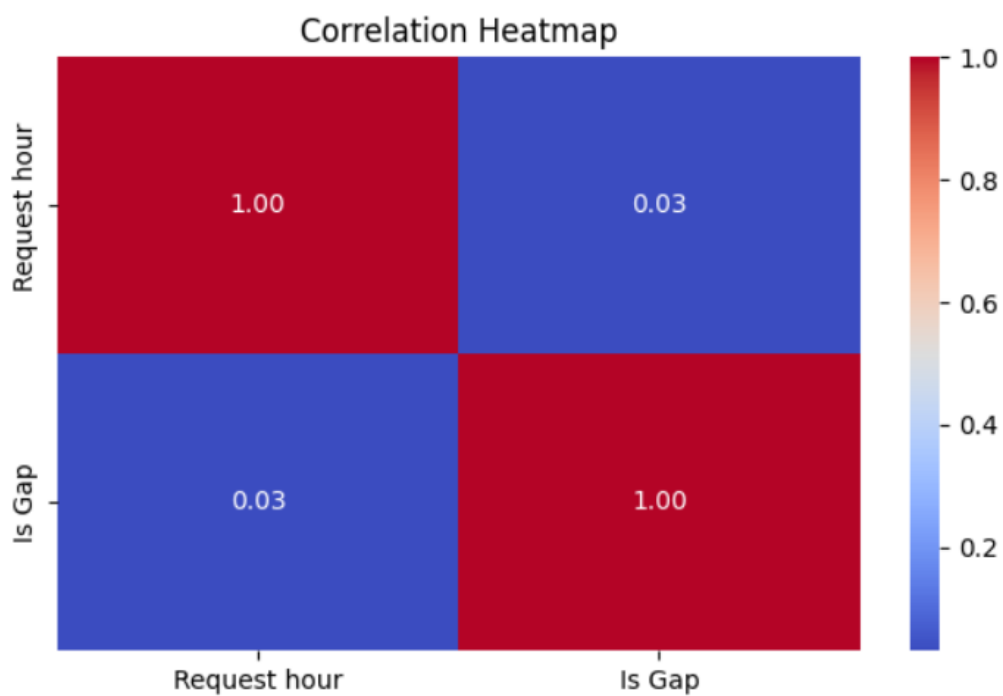


Requests by Hour & Status









Recommendations for the Business

1. Incentivize Drivers in Off-Peak Hours

Offer bonuses or surge pay for drivers accepting rides during Late Night and Early Morning.

2. Dynamic Fleet Allocation

Use historical data to predict demand patterns and shift idle drivers to critical zones like the Airport at night.

3. Reduce Cancellations

Investigate driver-side cancellation reasons (e.g., long-distance rides, ride value).

Penalize unnecessary cancellations and reward high-completion drivers.

4. Enhance Driver Availability at the Airport

Ensure more drivers are on standby at the Airport during critical night slots.

5. Improve Real-Time Monitoring

Deploy a live dashboard for Uber operations teams to track hourly request fulfillment and react quickly.

6. Customer Notification Systems

Notify users in high-gap areas (e.g., Airport at 2 AM) about potential wait times or suggest alternate transport.

7. Improve Mobile App Messaging

Show surge pricing or driver shortage warnings during peak demand to manage expectations.

8. Analyze Driver Behavior

Collect and study driver feedback during high cancellation slots to better understand challenges.

Excel Dashboard:

